

Exercise Exam Statistics AI 2020S

tags: Exam Statistics 2020S

A spam filter works by having users training the filter. It looks for patterns in the words in mails marked as spam by the user. For example, it may have learned that the word "win" appears in 30% of the mails marked as spam. It can be assumed that 0.9% of non-spams mail include the word "win" and 40% of all mails received by the user are spam.

- a) Calculate the total probability that the word "win" appears in a mail regardless the type of the mail. [15 points]
- b) Calculate the probability that a mail is not a spam mail, given the word "win" appears in it. [15 points]

a) 0.1254

b)

$$p(\text{win} \mid \text{no spam}) = 0.009$$

$$p(\text{win} \mid \text{spam}) = 0.3$$

$$P(\text{spam}) = 0.4$$

$$P(\text{no spam}) = 0.6$$

$$P(\text{no spam} \mid \text{win}) =$$

$$0.009 \cdot 0.6 / (0.009 \cdot 0.6 + 0.3 \cdot 0.4) = 0.043$$

Steel bolts are being manufactured to a specification which requires that their lengths be between 8.50 cm and 8.60 cm. A machine produces these bolts so that their lengths are normally distributed about a mean of 8.54 cm with a standard deviation of 0.02 cm.

- a) Calculate the percentage of bolts produced with a length of more than 8.54 cm. [10 points]
- b) Calculate the percentage of bolts produced that will not be within the required limits of length. [20 points]

a) 0.5 = 50%

b) Norml(8.54, 0.02)

- $P(8.50 < X < 8.60) = 0.9759$
- $1 - P(8.50 < X < 8.60) = 0.0241$

In the table below the body-mass-index (BMI) of a population of 50 people is given.

class limits $[e_{i-1}; e_i)$	frequency
[16 ; 20)	6
[20 ; 24)	20
[24 ; 28)	17
[28 ; 32)	7

- a) Calculate the arithmetic mean for the given data. [10 points]
- b) Calculate the median for the given data. [10 points]
- c) Above which value are 70% of the distribution mass? [20 points]

a) $1/N \cdot \sum (f_i \cdot e_{i-1} + e_i/2) \rightarrow 1200/50 \rightarrow \text{Mean} = 24$

b) 23.8

c) $x_{0.3} = e_{i-1} + d_i/f_i \cdot (N \cdot P - F_{i-1}) = 20 + 4/20 \cdot 50 \cdot 0.3 - 6 = 21.8$